

Hands-On, Experimental Computer Science

Reading about recursion won't teach you how to implement it on its own. This book includes many recursive code examples in both the Python and JavaScript programming languages for you to experiment with. If you're new to programming, you can read my book *Automate the Boring Stuff with Python*, 2nd edition (No Starch Press, 2019), or *Python Crash Course*, 2nd edition, by Eric Matthes (No Starch Press, 2019) for an introduction to both programming and the Python programming language.

I recommend stepping through these programs with a debugger. A *debugger* lets you execute programs one line at a time and inspect the state of the program along the way, allowing you to pinpoint where bugs occur. Chapter 11 of *Automate the Boring Stuff with Python*, 2nd edition, covers how to use the Python debugger and is free to read online at <https://automatetheboringstuff.com/2e/chapter11>.

The chapters in this book display the Python and JavaScript code examples together. The Python code is saved in a `.py` file, and the JavaScript code in an `.html` file (not a `.js` file). For example, take the following `hello.py` file:

```
print('Hello, world!')
```

And the following `hello.html` file:

```
<script type="text/javascript">
document.write("Hello, world!<br />");
</script>
```

The two code listings act as a Rosetta stone, describing programs that produce the same results in two different languages.
